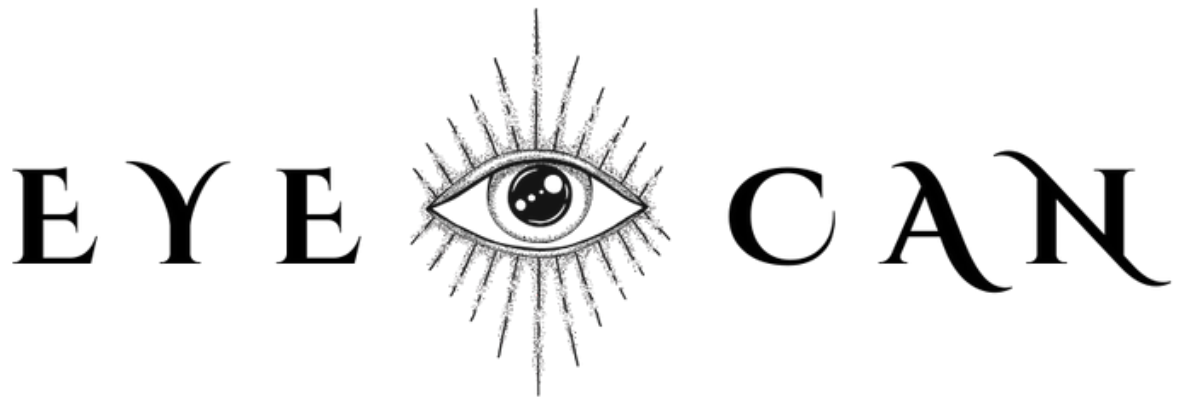




# **Eyecan – Eye-Tracking Smart Glasses Simulator for Communication**

Eyecan is an innovative smart glasses simulator that empowers individuals who cannot speak or move to communicate using only eye movements. It provides a seamless interactive experience, allowing users to express their thoughts and needs quickly and safely, restoring part of their independence in a unique technological and human-centered way.

## **Team Members:**

- Rama Al-Najjar – Software Engineering
- Maryam Al-Jalakh – Artificial Intelligence
- Aya Al-Ghraibeh – Cyber Security
- Najwa Al-Joulani – Artificial Intelligence

**From:** Faculty of information technology at Zarqa University

**Date :** October 2025

---

## Abstract

Eyecan is an innovative project that simulates smart glasses, enabling individuals who have lost the ability to speak or move due to wars, neurological injuries, diseases, or other harsh circumstances to communicate freely and easily using only eye movements. Many of these individuals once lived normal lives full of social interaction, mobility, and self-expression, but suddenly lost the sensory or physical abilities they once relied on. Eyecan aims to restore part of this lost life by providing a direct way to express thoughts and needs, giving users the opportunity to regain independence and dignity.

By integrating modern technology with a human-centered design, Eyecan offers a seamless interactive experience that mimics everyday life as closely as possible, becoming a true bridge connecting users to the world around them. It allows them to feel part of society again, capable of interaction, decision-making, and self-expression, as if they were living a normal life without physical constraints. Eyecan is more than a technical tool—it is a human-centered solution that empowers individuals to reclaim communication and social engagement, highlighting the crucial role of technology in improving the quality of life for those who have faced significant physical and psychological challenges.

---

## Problem Eyecan Solves

Many individuals around the world suddenly lose the ability to speak or move due to wars, neurological injuries, illnesses, or other harsh circumstances. This sudden loss prevents them from expressing their thoughts, needs, or interacting with their social environment, leading to feelings of isolation, loss of independence, and diminished dignity.

Although some assistive devices exist, most are either difficult to use, expensive, or fail to provide a natural interactive experience, especially for people with severe disabilities. This creates a significant gap between these individuals and the world around them, increasing their social isolation and limiting effective communication.

The Eyecan project aims to address this problem by providing an innovative solution: a smart glasses simulator based on eye-tracking that enables users to communicate easily and safely, restoring part of their independence and their ability to interact with the world.

---

## Objectives & Motivation

In a world where many individuals face significant physical and psychological challenges due to wars, injuries, or illnesses, some suddenly lose the ability to speak or hear, finding themselves cut off from their surroundings and unable to express their thoughts and needs. After the Gaza war, we witnessed many cases where people lost their speech or hearing due to physical and psychological trauma, and even after the conflict ended, they continue to struggle, unable to return to their normal lives.

We researched the topic deeply and considered how we could help them, aiming to be part of restoring their lives—not only for those affected by wars, but also for individuals who suddenly lost these senses due to accidents, illnesses, or psychological shocks. Eyecan was born from this profound human need: to serve as a bridge reconnecting people with the world around them, giving them the opportunity to interact and communicate freely, as if they were living a normal life again.

The project aims to empower these individuals to regain part of their independence and dignity by providing a seamless communication method using only eye movements, combining technology to restore a sense of life and lost senses. The goal is not merely technological; it is about creating a fully human experience, where the user feels heard, understood, and able to express themselves safely and freely. Eyecan becomes a window to the world, a means to restore social connections, reclaim a sense of worth and dignity, and live a life closer to the one they had before circumstances stripped them of their senses.

---

## System Architecture

The Eyecan system is designed to provide a fully integrated communication experience using a virtual smart glasses web application, based on several interconnected modules:

### 1. User Interface (UI):

- Page 1: Glasses catalog displaying general product information.
- Page 2: Interactive prototype allowing users to select categories and options via eye movement- Glasses settings: allow users to choose male or female voice, adjust gaze duration (1-3 seconds), train the system to their eyes, and turn the glasses on/off.

### 2. Eye-Tracking Module:

- Captures user eye movements accurately and converts gaze on interface elements into digital commands.
- Includes training mode to calibrate the glasses for each user and improve tracking accuracy.

### 3. Context & Selection Engine:

- Determines the category and specific choice the user is focusing on.
- Displays appropriate options within each category, e.g., “Good morning,” “I want to eat,” “I want to drink,” etc.

### 4. Text-to-Speech Module:

- Converts user selections into audible voice output according to the chosen settings (male or female voice).

---

## System Architecture

### 5. Speech-to-Text & AI Suggestions:

- Captures surrounding sounds and converts them into text displayed on the screen.
- AI suggests suitable responses which the user can select, then converted into voice output.

### 6. Database:

- Stores user preferences, glasses settings, eye-tracking calibration data, phrases, and options.
- Supports multiple languages and customization for each user.

### 7. Power Control:

- Turns the glasses on/off based on the duration of gaze on the power button, ensuring user comfort and ease of use.

All components work together seamlessly, allowing the user to:

- Select categories and phrases using eye movements.
- Convert selections into audible speech.
- Listen to the environment and respond via AI suggestions.
- Customize the glasses settings for comfort.
- Experience a safe and comprehensive communication system that closely mimics natural life.

---

## Scenario

Imagine Sara, who lost her ability to speak after a severe accident, feeling isolated from the world around her. Today, she enters the Eyecan prototype page via her web browser.

### 1. First Page – Catalog:

- Sara sees an introduction to the smart glasses and its features, learning that it can convert eye movements into audible speech.

### 2. Second Page – Interactive Prototype:

- Upon entry, several categories appear: Greetings, Food & Drink, Questions, Daily Needs.
- Sara focuses her eyes on the "Greetings" category, then a box labeled "Good Morning" appears. By maintaining her gaze for two seconds, the text converts into audible voice that the people around her can hear immediately.

### 3. Listening & Responding:

- Sara wants to hear what others are saying, so the microphone icon appears. After two seconds, the system captures surrounding sounds and converts them into text on the screen.
- AI suggests suitable responses such as "Good evening" or "How are you?". Sara selects the appropriate response, which is then voiced to everyone nearby.

### 4. Settings:

- Sara can choose a male or female voice, adjust gaze duration, or train the system for her eyes to enhance accuracy.
- She can turn the glasses on/off by looking at the power icon for a set duration.

In this way, Sara is able to \*\*communicate freely, express her needs, and participate in conversations\*\*, as if she had not lost any abilities, regaining part of her normal life.

Eyecan provides each user with a \*\*human-centered, easy, and safe experience\*\*, enabling natural communication despite the loss of speech or hearing.

---

## Implementation

The Eyecan system is developed using an integrated architecture leveraging modern technologies for frontend, backend, and APIs, ensuring a seamless and effective communication experience.

### 1. Frontend:

- Built using React with TypeScript to provide a responsive and interactive user interface.
- Includes two main pages:
  - Page 1: Glasses catalog, displaying features and information.
  - Page 2: Interactive prototype, where users select categories and options via eye-like movement simulated with a mouse.
- UI supports multiple languages and interactive display of buttons, boxes, microphone, and power icons.

### 2. Backend:

- Developed using Python to create a flexible and scalable system.
- Utilizes Google APIs to provide core functionalities:
  - Text-to-Speech (TTS) to convert selected text into audible voice.
  - Speech-to-Text (STT) to convert surrounding sounds into text on the screen.
  - AI Suggestion API from Gemini to provide intelligent response suggestions.
- Firebase is used as the database to store user preferences, glasses settings, Eye-Tracking training data, phrases and categories, and interaction logs.

### 3. Eye-Tracking (Simulation):

- In the prototype, mouse movement is used to simulate eye tracking.
- The mouse cursor determines the category and option the user focuses on.

With this architecture, Eyecan ensures a fully integrated communication experience where data flows from simulated eye movement to context analysis, text-to-speech conversion, and environment response, seamlessly integrating AI to provide intelligent suggestions and a rich interactive experience.



---

## Testing & Evaluation

The Eyecan system was tested at multiple levels to ensure efficient operation and provide a seamless and safe communication experience:

### 1. Frontend Testing:

- Verified the responsiveness of all buttons, boxes, microphone, and power icons.
- Tested the category and option selection experience using mouse simulation to ensure accurate focus recognition.
- Ensured correct display of texts across multiple languages.

### 2. Backend Testing:

- Validated proper operation of Python APIs with Google APIs (TTS, STT, AI Suggestions).
- Tested Firebase integration to ensure fast and accurate data storage and retrieval.
- Checked system stability under multiple users and simultaneous interactions.

### 3. Integration Testing:

- Tested the data flow between Frontend and Backend, including:
  - Sending simulated eye-tracking data.
  - Receiving AI-generated voice and text suggestions.
  - Updating the database with user selections and preferences.
- Ensured all components work cohesively without errors.

### 4. User Evaluation:

- Conducted trial sessions with a group of users to assess usability and system effectiveness.
- Measured speed of option selection, voice clarity, and AI suggestion accuracy.
- Collected user feedback on navigation and communication experience to ensure a human-centered, natural interaction.

Through these tests, Eyecan has been confirmed to provide a safe, accurate, and usable communication experience, focusing on **\*\*empowering users to express themselves and communicate freely\*\***.

---

## EyeCan Future Work

The future of the EyeCan project opens numerous opportunities for system development and enhancing user experience, while maintaining its human-centered approach:

### 1. Support for Real Devices:

- Develop a version that works on actual smart glasses instead of web/mouse simulation, providing a more realistic and accurate eye-tracking experience.

### 2. Improved Eye-Tracking Technology:

- Integrate advanced eye-tracking devices to reduce response time and increase focus recognition accuracy.
- Develop intelligent algorithms to adapt to individual user eye differences and ensure eye comfort for extended use.

### 3. Expanded User Scope:

- Add a mode supporting individuals who have been unable to hear or speak since birth.
- Include sign language as an alternative to text or voice communication, allowing these users to express themselves freely using visual gestures.

### 4. Enhanced AI Capabilities:

- AI suggestions become more contextual and personalized based on user interaction history.
- Integrate machine learning to analyze individual communication patterns and provide customized options for each user.
- Enable intelligent response suggestions based on the user's psychological state, where the glasses can detect biological indicators (e.g., heart rate, stress signals) to provide supportive and appropriate responses.

---

## EyeCan Future Work

### 5. Voice Cloning:

- Ability to replicate the user's original voice if they were previously able to speak and hear.
- Allows people around the user to hear their original voice, enhancing human connection and personal touch in communication.

### 6. Advanced Audio and Visual Experience:

- Add the ability to convert selected texts into interactive images or symbols to assist users with severe disabilities.
- Enhance voice quality to be more natural and realistic, with spatial audio support for an immersive experience.

### 7. Integration with Other Technologies:

- Integrate Eyecan with mobile applications or VR/AR systems for seamless and accessible communication.
- Enable direct sharing of messages and interactions with family, friends, or specialists.

Future work aims to make Eyecan a **\*\*comprehensive, intelligent, and adaptable communication tool\*\***, focusing on empowering individuals who have lost their natural abilities to express themselves and interact with society, whether due to temporary or permanent hearing and speech loss, while supporting their psychological and emotional well-being through context-aware AI suggestions.

---

## Conclusion

The Eyecan project presents an innovative communication model for individuals who have lost the ability to speak or hear, whether due to wars, trauma, illness, or from birth. By integrating eye-tracking technology, artificial intelligence, text-to-speech and speech-to-text conversion, and sign language support, users can freely express themselves and regain a sense of social independence.

Eyecan is not just a technological tool; it is a **\*\*human bridge connecting individuals to the world around them\*\***, taking into account the psychological and emotional needs of each user. The project demonstrates how technology can empower people and enhance their quality of life, providing a natural, seamless, and personalized communication experience.

Although the current version is a web-based simulation, the project opens vast possibilities for the future, including real smart glasses, sign language integration, original voice cloning, and AI suggestions tailored to the user's psychological state, making it a comprehensive and effective communication tool for those in need.

---

## References and Technologies

The following tools and technologies were used in the development of the Eyecan project:

### 1. Frontend:

- React.js with TypeScript for building an interactive user interface.
- HTML5 and CSS3 for page structure and styling.
- Tailwind CSS for modern and responsive UI design.

### 2. Backend:

- Python for server-side development and backend logic.
- Google APIs:
  - Text-to-Speech (TTS) for converting text into voice.
  - Speech-to-Text (STT) for converting sounds into text.
  - AI Suggestion API from Gemini for intelligent response suggestions.
- Firebase for data storage, user settings, and interaction logs.

### 3. Eye-Tracking Simulation:

- Mouse used as a simulation for eye movement in the prototype.
- Focus timer and training setup to accurately mimic eye tracking.

### 4. Artificial Intelligence Tools:

- Machine Learning to analyze individual communication patterns.
- AI-powered suggestions to provide context-aware and psychologically sensitive responses.
- Note: Some external AI tools were used to facilitate intelligent functionalities, fully documented to ensure transparency and honesty in the development process.

### 5. Additional Tools:

- Screen recording and video tools for documenting testing sessions.
- Graphic design tools for interface elements and icons.